2	**(a)**		Substitute $x = 40$, $h = 17 + 15\cos 240 = 9.5\,\text{m}$	**B1**	**1.1b**	cao
				[1]		
2	**(b)**		Maximum when $6x = 0$ or 360 so $x = 0, 60$	**M1**	**2.1**	Attempt to use the period of the $\cos 6x$ function. Must include a reference to either 60 or 240 or a sketch illustrating the x-direction stretch. Allow if wrong conclusion reached. Do not allow for an argument based on mechanics principles alone
			but the model is only valid for $0 \le x \le 40$ So Tom's argument is invalid.	**E1**	**2.3**	clear argument
				[2]		